

Main Project Report

Date: 25/08/2026

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Dataset: HealthConnect Appointment Dataset

PROJECT & RESOURCE REVIEW

1. Problem to address

HealthConnect Clinic is experiencing operational challenges due to patients missing scheduled appointments and they have difficulty understanding the factors associated with appointment no-shows

2. Role in HealthConnect Project

As a **Data Analytics** intern, my responsibility is to understand factors associated with appointment no-shows, identify patterns in patient behavior and appointment characteristics, provide actionable insights to help reduce missed appointments. Lastly support decision making by establishing meaningful KPI's and dashboards.

3. Resource to use

HealthConnect_Appointment_Data.csv: primary source of data

HealthConnect_Data_Dictionary.xlsx: understand variable meanings and data types

Week4 Assignment Brief: Guide with the scope and deliverables

Python: To understand the dataset e.g. data profiling

Power BI: To understand the dataset and visualization

4. Initial approach to the project

Data Understanding

- Import all necessary python libraries and import my dataset in python
- Review data dictionary to understand variable types and dictionaries
- Check total records and fields

Data Quality Assessment

- Identify missing values, duplicates, data types, validated calculated fields

Exploratory Data Analysis

- Identify patterns in no-show behavior
- Formulate business questions
- Identify KPI cards

Dashboard Development

- Create KPI measures
- Design dashboard layout and build interactive dashboard
- Add filters for deep analysis

TRACK SPECIFIC ASSIGNMENT

1. Dataset Overview

The primary dataset for analysis is HealthConnect_Appointment_Data.csv which contains 5000 appointment records and 18 fields from HealthConnect Clinic, spanning dates from 2024 to 2026.

Key Attributes

Patient Demographics: Patient_Id, Gender, Age, AgeGroup

Appointment Details: appointment_Id, appointment_type, appointment_date, appointment_day, appointment_time, appointment_outcome

Booking Details: booking_date, booking_lead_days

Appointment History: previous_appointments, previous_no_shows.

Reminder & Logistics information: reminder_sent, reminder_channel, distance_to_clinic_km, waiting_time_minutes.

Target Variable: appointment_outcome(Attended, no-show, cancelled)

2. Data Quality Assessment

The analysis of my dataset reveals several data quality considerations that will need to be handled during the analysis phase.

1. Missing Values

The dataset has several missing values which can affect the analysis if not addressed properly.

Distance_to_clinic_km: has 90 missing values.

Waiting_time_minuted: has 60 missing values.

Reminder_channel: has 1366 missing values.

Methodology

Used Python pd.isnull().sum() to get the count of all missing values

Also used Power BI power query's column quality, distribution and profile features to identify missing values, which revealed that the null values which were counted by python are not actually null because it has another option being "none" which python mistaken for null. And this was already described in the data dictionary.

Action

We will need to decide strategy for handling these missing values, options include assigning 0, excluding these records or computing them.

2. Data Type Inconsistency

Most fields have the right data type except for 2 being booking_date & appointment_date, which are listed as text.

Action

Dates will be converted to date time format to allow for time-series analysis

3. Data Formatting

We have a field booking_lead_days which represents the time difference between booking and appointment, we need to validate the existing values and check for accuracy.

Action

This field will be re-calculated using appointment_date - booking_lead_days to ensure consistency and possibly identify any data entry errors

3. Relevant Business Questions

The main question of the project is “How can HealthConnect Clinic use data to reduce missed appointments and improve patient support experience”. To be able to answer this we need to dive deep into the following questions that can help us answer our main questions:

- What demographic factors are most strongly associated with higher no-show rates?
- Are there specific days of the week, times of day, or appointment types that have higher no-show rates?
- What is the relationship between a patient's appointment history and their likelihood of missing a future appointment?
- Which months have high bookings but low turn up?
- How does reminder & logistic information affect no-show rates?
- How do 'No-Show' appointments compare to 'Attended' and 'Cancelled' appointments across all other factors?
- Which reminder channel is used most and works more efficiently?

4. Proposed KPI's

KPI's are mostly used to measure progress towards the goal of analysis, identify problems that need attention, enable decision makers to make decisions at glance and track performance overtime. Based on the core business problem and my business problems I have identified the following KPI's:

Total Bookings: Count of all appointments booked during the reporting period. Answers the business question “Which month as high booking dates but low turn up”? When paired with month filter

Total Attended Appointment: Count of appointments with outcome = “Attended” and address the business question “Does demographic affect appointment outcome”? when paired with demographic filters

Total no-show: Count of all appointments with outcome = “No-show,” related to the question “Are there specific days of the week, times of day, or appointment types that have higher no-show rates?”

No-show rate: $(\text{Total no-show} / \text{Total Bookings}) * 100$, answer most question based on the filter applied for example “How does reminder & logistic information affect no-show rates?” When we apply the reminder channel filter

Show rate: $(\text{Total show} / \text{Total Bookings}) * 100$, answer most questions above based on the filter that will be applied e.g. “Which reminder channel is used most and works efficiently?” when we apply the reminder channel filter.

Previous no show: Count of patients with a history of previous no-shows. Answer the following question “Does patient history influence appointment outcome”?

5. Initial Analysis Approach

Data Preparation & Cleaning

- From my python analysis I’ve identified missing values which will be handled in week 5 by importing the dataset and addressing missing values by either assigning all null values 0 since they are numerical and not calculated or assigning them the median values.
- Converting date columns to proper datetime format
- Categorizing my booking_lead_days and previous_no_shows for easier analysis
- Validate calculated fields such as booking_lead_days

Hypothesis Testing & Insight Generation

I’ve already listed my objectives, i.e., business questions to address, from here I will formalize insights from the analysis phase and areas for improvement.

Reporting & Visualization

I will now summarize all my findings into a clear and concise report using Power BI by creating a dashboard with proposed KPI’s and filters for deep

ASSUMPTIONS, LIMITATIONS, RISKS AND DEPENDENCIES

1. Assumptions

- The “none” value in reminder channel correctly indicates that no reminder was sent, rather than entry error
- The calculated fields age_group and booking_lead_days are accurate and derived consistently
- The data dictionary provided is accurate and complete

2. Limitations

- The dataset is entirely fictional and may not reflect real-world healthcare data patterns
- The absence of socio-economic variables limits the ability to understand all drivers of no-shows
- No information about appointment rescheduling is available, which could impact the interpretation of cancellation data

- The reminder data shows percentage of reminders_sent, it's not known if the message content was the same which could influence effectiveness.

3. Risks

- Inconsistent handling of missing values could lead to misleading conclusions.
- It is unclear if waiting_time_minutes is an actual historical wait time or an estimate.
- Some values are missing which could introduce bias if the missing data is not random.